

Lab 5 - Friction

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Learning objectives

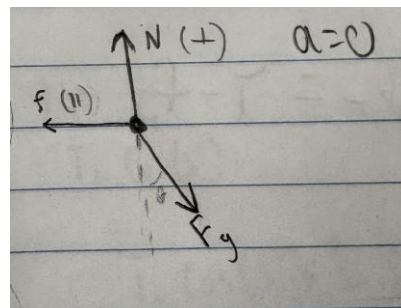
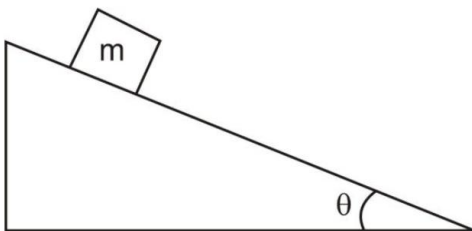
- Use a spreadsheet to record and analyze data
- Draw a free-body diagram
- Apply Newton's first law of motion
- Understand friction

Equipment

- Capstone data/sensor interface
- Inclined plane apparatus (ramp)
- Wooden block of mass m with hook
- Paper (on apparatus and for tabletop)
- Additional masses: 250g, 500g, 750g, 1.00kg, 1.25g, 1.50g
- Force sensor with hook
- String (~0.5 m in length)
- Google sheet for data collection and analysis

Part 1 - Friction on an inclined plane (ramp)

A block of mass m is at rest on a ramp. The ramp makes an angle of incline θ with the horizontal.



Prelab:

1. Draw a free-body diagram of m to the right of the figure and label the forces acting. Use Newton's second law to express the frictional force $\mathbf{F}_f$ and the normal force $\mathbf{F}_n$ in terms of the variables $\mathbf{g}$, $\mathbf{m}$, and θ . (Hint: choose orthogonal axes that are parallel ($\parallel$) and perpendicular ($\perp$) to the ramp and assume that the block is at rest).

$$F_n = mg \cos \theta$$

$$F_f = mg \sin \theta$$

2. Solve for F_f and F_n when $\theta = 0$. Is this what you expect? Explain using Newton's laws.

$F_n = mg \cos(0) = mg$ When there is no slope the normal force is perpendicular to the pull of gravity causing an equal and opposite force. (Newton's third law)

$F_f = mg \sin(0) = 0$ When there is no slope for gravity to pull the block down there is no need for an equal force (friction) to oppose the motion. (Newton's third law)

$$F_f = mg(\mu_s)$$

$$F_n = (F_f / \mu_s)$$

When the block is at rest, static friction is the frictional force that is acting. The model for simple static friction is the equation $F_f \leq \mu_s F_n$, where μ_s is the coefficient of static friction.

As the angle of incline θ increases, the frictional force grows until it reaches its maximum value. Then $F_f = \mu_s F_n$. Greater than this angle, the static friction cannot get any larger and the block slips. This angle is called the "critical angle" θ_c .

3. Substitute $F_f = \mu_s F_n$ into the equations you derived for F_f and F_n previously. Solve the equations for μ_s when $\theta = \theta_c$. You should now have an expression for μ_s that depends only on θ_c . Which trig function do you recognize?

$$\mu_s = (\sin(\theta_c) / \cos(\theta_c))$$

$$\mu_s = \tan(\theta_c)$$

Measurements:

Measure the mass of the block of mass m: 129.67 g. Convert to SI units: 0.12967 kg.

Make sure there is paper on the inclined plane apparatus. **Be careful not to touch the paper with your fingers.** Doing so may add oil to the paper and change its coefficient of static friction!

Place the block of mass m on the inclined plane with the wide side facing down. Slowly and gently lift the incline until the block starts to move. Determine a range of angles that you are sure the critical angle falls within. (Hint: slide another wooden block under the ramp, back and forth for better control).

Record this as $20.2 < \theta_c < 21.8$.

Use this range of angles to determine a range of coefficients that μ_s falls within.

Record this as $0.3679 < \mu_s < 0.4000$

This represents the coefficient of static friction for **wood** on **paper**.

If there is time after Part 2 (error bars)

Record a column of 3-4 measurements of θ_c in a [spreadsheet](#). For every measurement, calculate and record μ_s . (Each group member should measure, calculate and record once).

Use googlesheet functions to determine the mean and standard deviation for θ_c and μ_s .

Calculate the standard error sigma for each and **report** as “measurement = mean $\pm$ 2σ .”

(Refer to [Basic data analysis-2A](#) in Module 0 if needed).

θ_c = mean: 21.15

STD: 0.7187952884

Standard error: 21.15 $\pm$ 0.7188

μ_s = mean: 0.386925

STD: 0.01440610403

Standard error: 0.386925 $\pm$ 0.0144

Part 2 - Friction on a Horizontal Surface

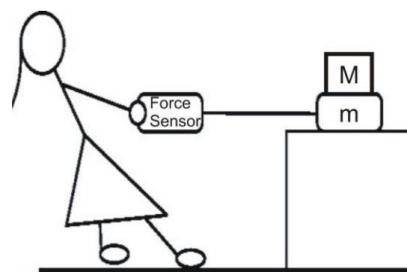
Setup:

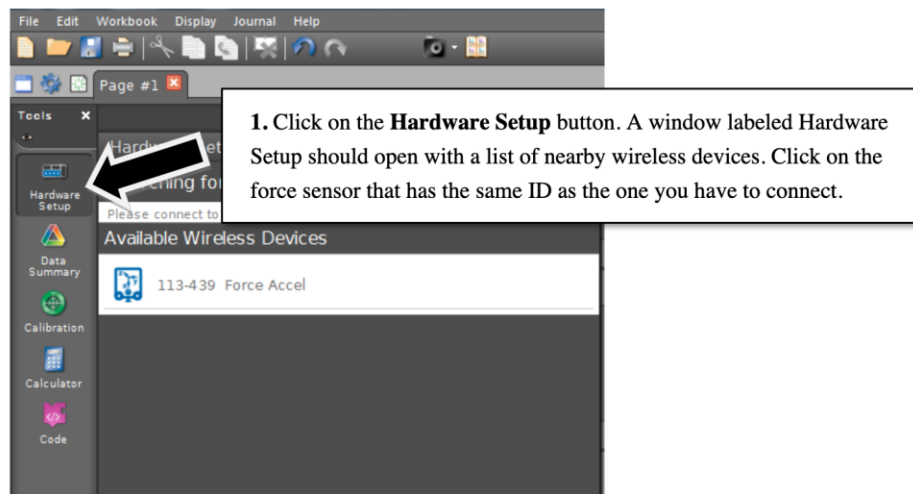
1. Tie a string to the hook attached to the wooden block **m** and place a 250-gram mass **M** on top of the wooden block. Tie the other end of the string to the hook on the force sensor.

2. Tape a length of paper roll to the tabletop, enough so that you can pull the wooden block across the paper at a constant speed.

3. Press the power button located on the side of the wireless force sensor to turn it on. When the Bluetooth indicator light is flashing red, it means the sensor is waiting to be connected to a laptop.

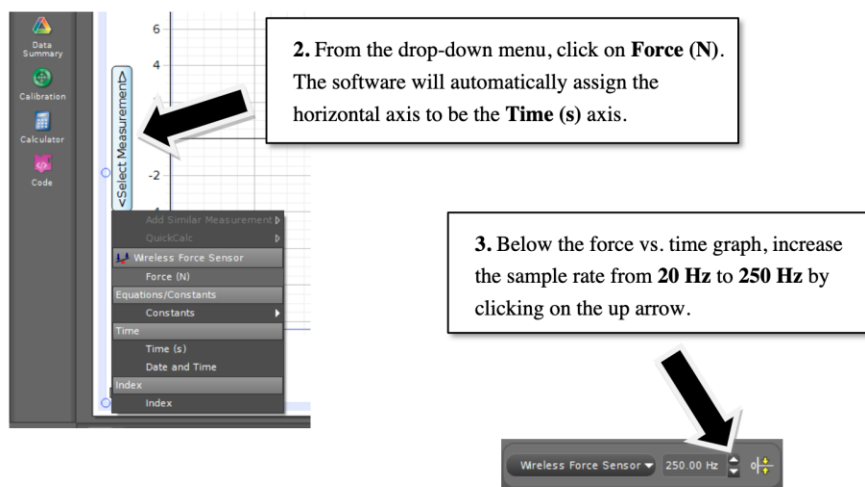
4. Open the Capstone software on the Dell laptop. Click on the Hardware Setup button located on the left side of the screen. This will open a window displaying nearby wireless devices.





5. After connecting the force sensor, click on the sliders to disable Wireless Acceleration Sensor and Wireless Gyro Sensor. Only the Force Sensor will be used in this experiment. Click on the Hardware Setup button again to close the Hardware Setup window.

6. Double-click on the Graph icon to open a graph. On the vertical axis, click on the <Select Measurement> button.



7. **Zero the force sensor** by removing any tension in the string connected to the force sensor hook and click on the icon with the "0" and two yellow arrows.

Data collection:

Start collecting data by clicking the Record button. Note that when the force sensor hook is pulled, it measures force to be negative. When it is pushed, it measures positive values. You only need to record positive values or the magnitude of the force measured.

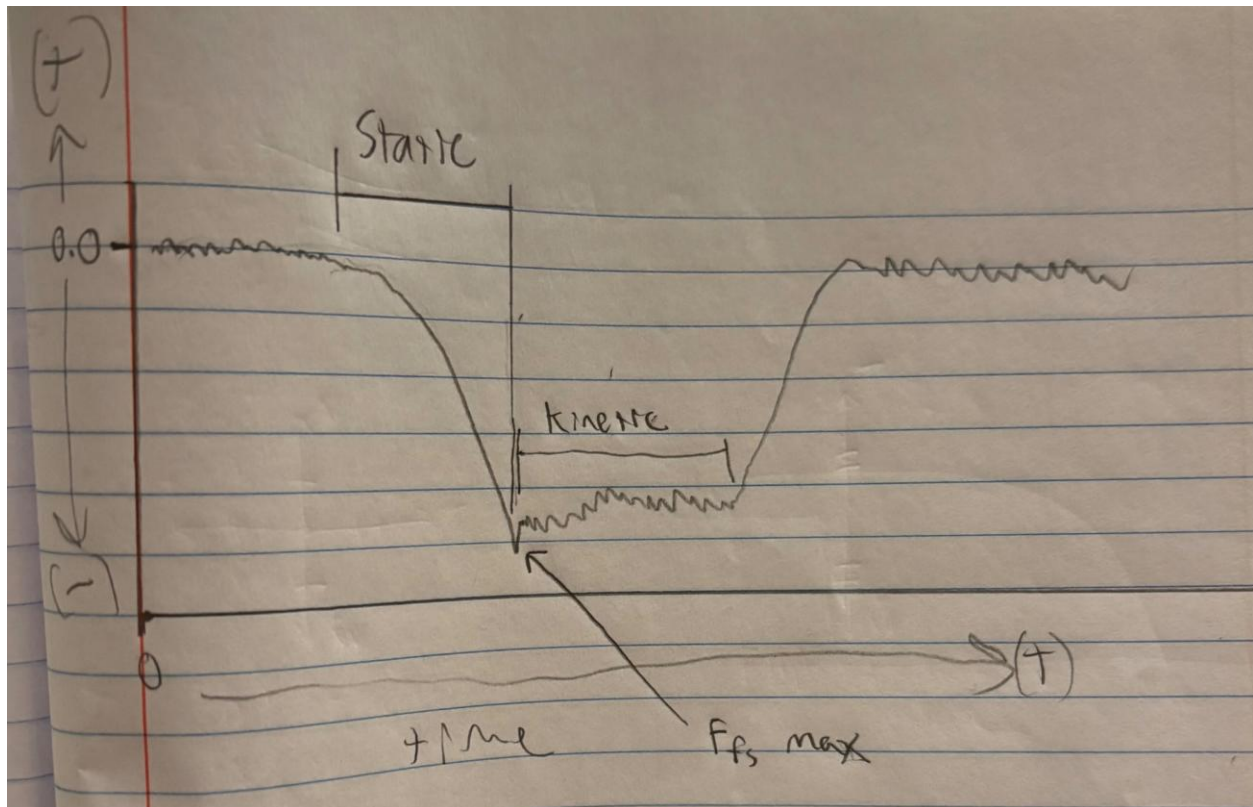
Pull the block horizontally along the paper with a slowly increasing force until the block just starts to move. Once the block is in motion, try your best to pull at a constant speed. You may need to do a few practice pulls to get it right!

Sketch the general shape of the force vs. time graph recorded in Capstone and label it as follows, then insert a picture:

Label the region of the plot where the block was at rest with static friction acting as "STATIC."

Label the point of maximum static friction as " F_{fs_max} ."

Label the region of the plot where the block was moving at constant speed as "KINETIC."



Is the maximum static friction greater than the kinetic friction? Describe how your plot answers this question.

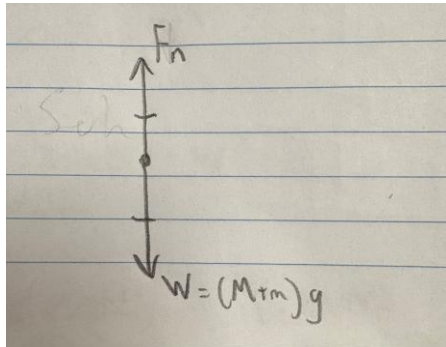
The maximum friction is greater than the kinetic friction. The plot shows this because the highest point of force is the exact moment before the block starts moving. Once the block had started to move the values of force lowered.

Is the kinetic friction a constant value? Describe how your plot answers this question. Kinetic friction is a constant magnitude when human error is accounted for in our graph. When looking at the graph, excluding all spikes and dips caused by jittery human motion, the values all roughly fall on the same flat line showing that the force is constant.

Let m be the mass of the wooden block and M be the mass placed on top of the wooden block. Determine the normal force F_n in newtons between the block and the table. Explain your reasoning behind how you got this value (Hint: draw a FBD for the system of $m + M$).

$$F_n = (M + m)g$$

This is because since there is no slope the normal force is perpendicular to the direction of gravity. This means any force of weight downwards is equivalent to the normal force as stated by Newton's third law.



Enter 7 values for M in your spreadsheet starting with 0kg up to 1.5kg in increments of 250g. Record a plot of Force vs. time when pulling the wooden block with the following values of mass M placed on top: 250, grams, 500 grams, 750 grams, 1 kg, 1.25kg, and 1.5 kg.

For each of these plots, record 3 forces in your spreadsheet: the normal force (F_n) between the block and the table, the maximum static frictional force (F_{fs_max}), and the kinetic frictional force (F_{fk}). If the kinetic frictional force varies, choose what you think is a reasonable representative value. Record the force values that you measure as positive. Check units!

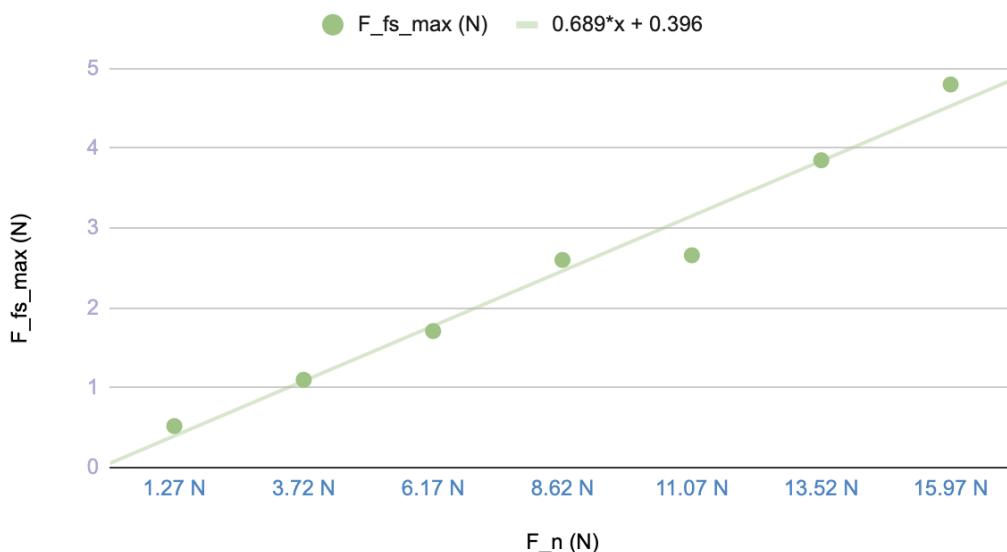
Insert a screenshot of your data table for Parts 1 and 2.

Part 1			Part 2					
m = 0.12967 Kg			m = 0.12967 kg.					
measurement #	$\theta_c (^\circ)$	μ_s		M (kg)	m + M (kg)	F _n (N)	F _{fs_max} (N)	F _{fk} (N)
1	20.2	0.3679		0	0.130 kg	1.27 N	0.52	0.4
2	21	0.3839		0.25 kg	0.38 kg	3.72 N	1.1	1
3	21.6	0.3959		0.5 kg	0.63 kg	6.17 N	1.71	1.57
4	21.8	0.4		0.75 kg	0.88 kg	8.62 N	2.6	2.11
mean	21.15	0.386925		1 kg	1.13 kg	11.07 N	2.66	2.57
stdev	0.7187952 884	0.0144061 0403		1.25 kg	1.38 kg	13.52 N	3.85	2.82
st error σ	0.3593976 442	0.0072030 52		1.5 kg	1.63 kg	15.97 N	4.08	3.36

Graphing results to determine μ_s and μ_k

1. Make a scatter plot of maximum static frictional force vs. normal force.

F_{fs_max} (N) vs. F_n (N)

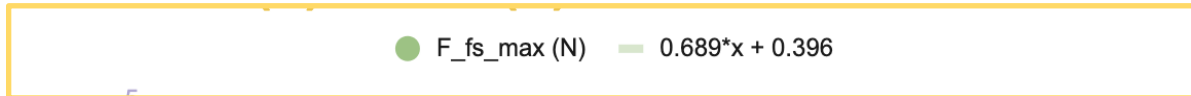


Does the shape of the plot match the model for simple static friction? (Hint: remember the equation highlighted in Part 1).

Yes, it does, as it is linear and positive, which matches the constant for static friction.

Do a best fit to the scatter plot to determine the coefficient of static friction and insert a screenshot of your fit result.

Yes,



Compare this value of the coefficient of static friction to what you obtained in Part 1. (Hint: does it fall within the range you determined?. Can you calculate a %-difference?)

Measured value - theoretical value / theoretical value = % difference.

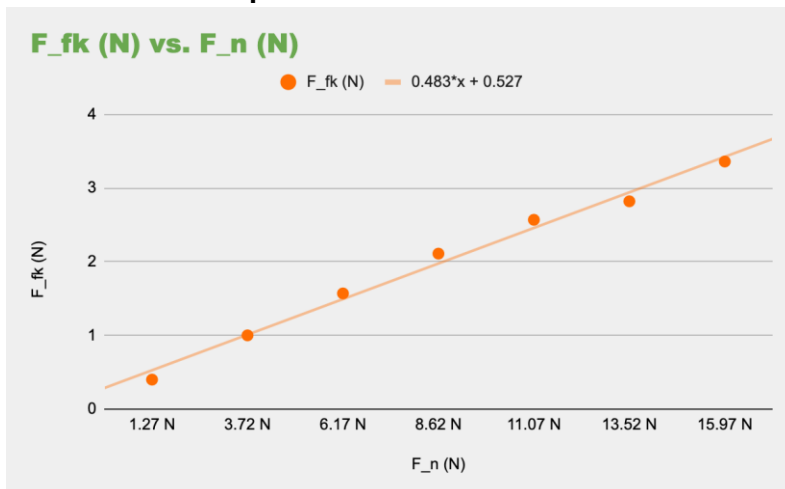
$$0.386925 - 0.689 / 0.689 \times 100 \% = 43.84 \%$$

- This high % error difference is because we used two different sheets of paper between each part, the one used for the second part was significantly rougher.

Compare the quality of the results. Which method of determining μ_s is more trustworthy?

The graph is more trustworthy in determining static friction because it has multiple trials and takes the average for a more exact result.

2. Make a scatter plot of kinetic frictional force vs. normal force.



Does the shape of the plot match the model for simple kinetic friction? (Hint: write the equation).

The model of simple kinetic friction is the force of friction and the force of push or pull is equal but the motion is in one direction thus meaning the object is being moved even though the friction and pull/push force is equal.

Yes, the shape of the plot does match the model since the force of friction is increasing along with the force of pull/push force. We can see the plot is directly proportional.

Do a best fit to the scatter plot to determine the coefficient of kinetic friction and insert a screenshot of your fit result.



Compare your graph estimates of μ_k and μ_s . Do the results support the model of simple friction, that an object in motion is easier to move than an object at rest?

The graph estimates of kinetic friction and static friction do support the model of simple friction, as kinetic friction was lower than static friction. This supports the idea that static friction force is larger than kinetic friction.

Part 1

m =

measurement #	$\theta_c (^\circ)$	μs
1	20.2	0.3679
2	21	0.3839
3	21.6	0.3959
4	21.8	0.4
mean	21.15	0.386925
stdev	0.718795288	0.014406104
st error σ	0.359397644	0.007203052

Part 2

m = 0.12967 kg.

M (kg)	m + M (kg)	F _n (N)	F _{fs_max} (N)	F _{fk} (N)
0	0.130kg	1.27 N	0.52	0.4
0.25 kg	0.38 kg	3.72 N	1.1	1
0.5 kg	0.63 kg	6.17 N	1.71	1.57
0.75 kg	0.88 kg	8.62 N	2.6	2.11
1 kg	1.13 kg	11.07 N	2.66	2.57
1.25 kg	1.38 kg	13.52 N	3.85	2.82
1.5 kg	1.63 kg	15.97 N	4.8	3.36

	F _{fs_max} (N)	F _n (N)
	0.52	1.27 N
	1.1	3.72 N
	1.71	6.17 N
	2.6	8.62 N
	2.66	11.07 N
	3.85	13.52 N
	4.8	15.97 N

F _n (N)	F _{fk} (N)
1.27 N	0.4
3.72 N	1
6.17 N	1.57
8.62 N	2.11
11.07 N	2.57
13.52 N	2.82
15.97 N	3.36

43.84 % (
19.89 % (static)

